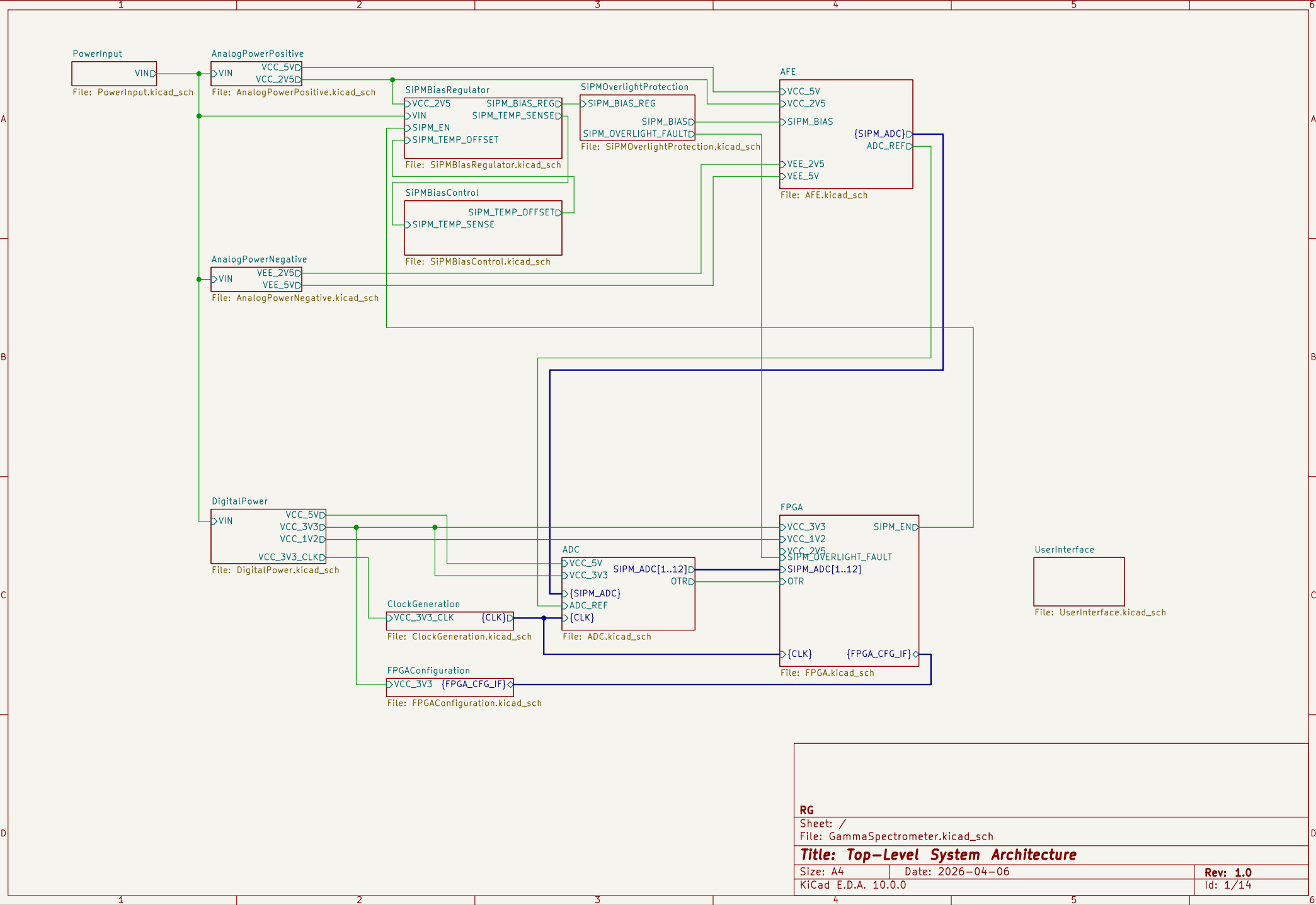
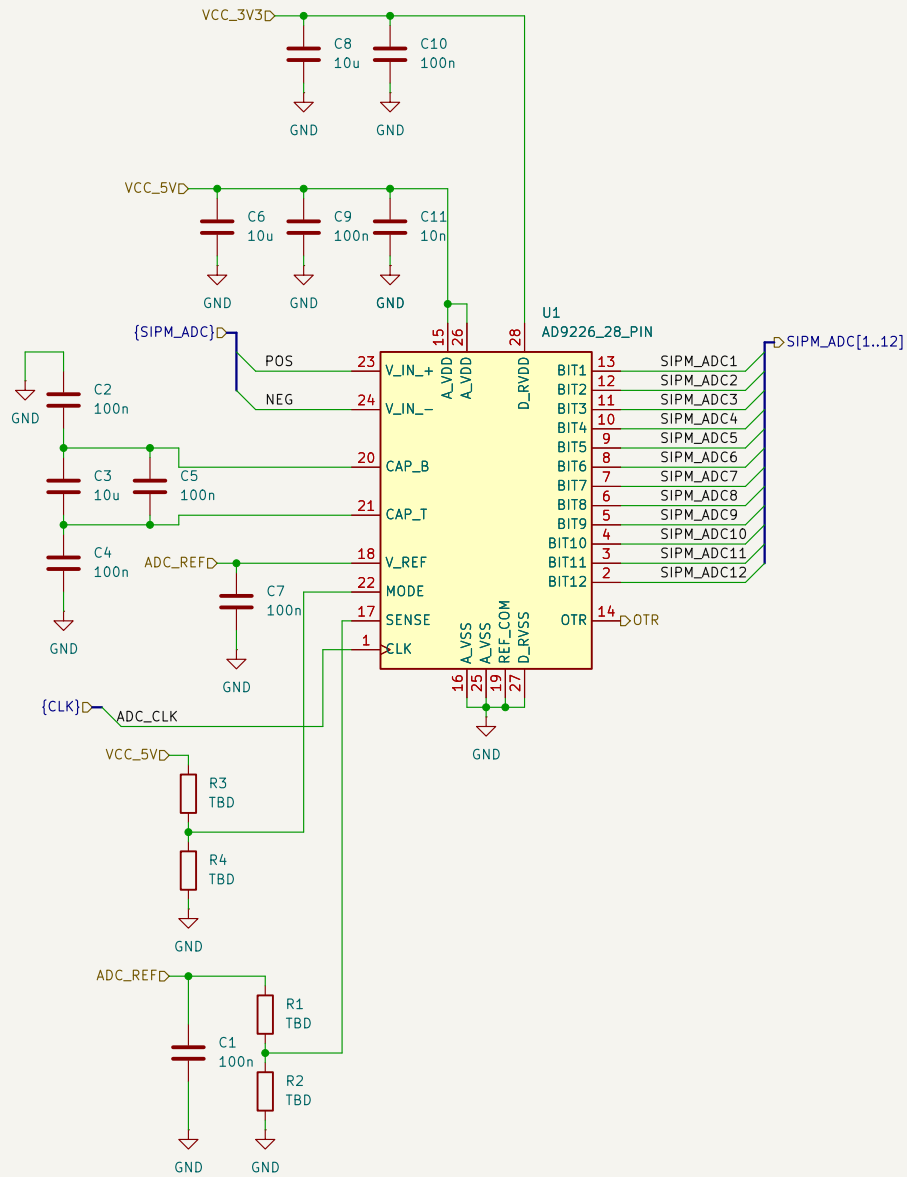






RG

Sheet: /ADC/
File: ADC.kicad_sch

Title: 12-bit 65MSPS Analog-to-Digital Converter

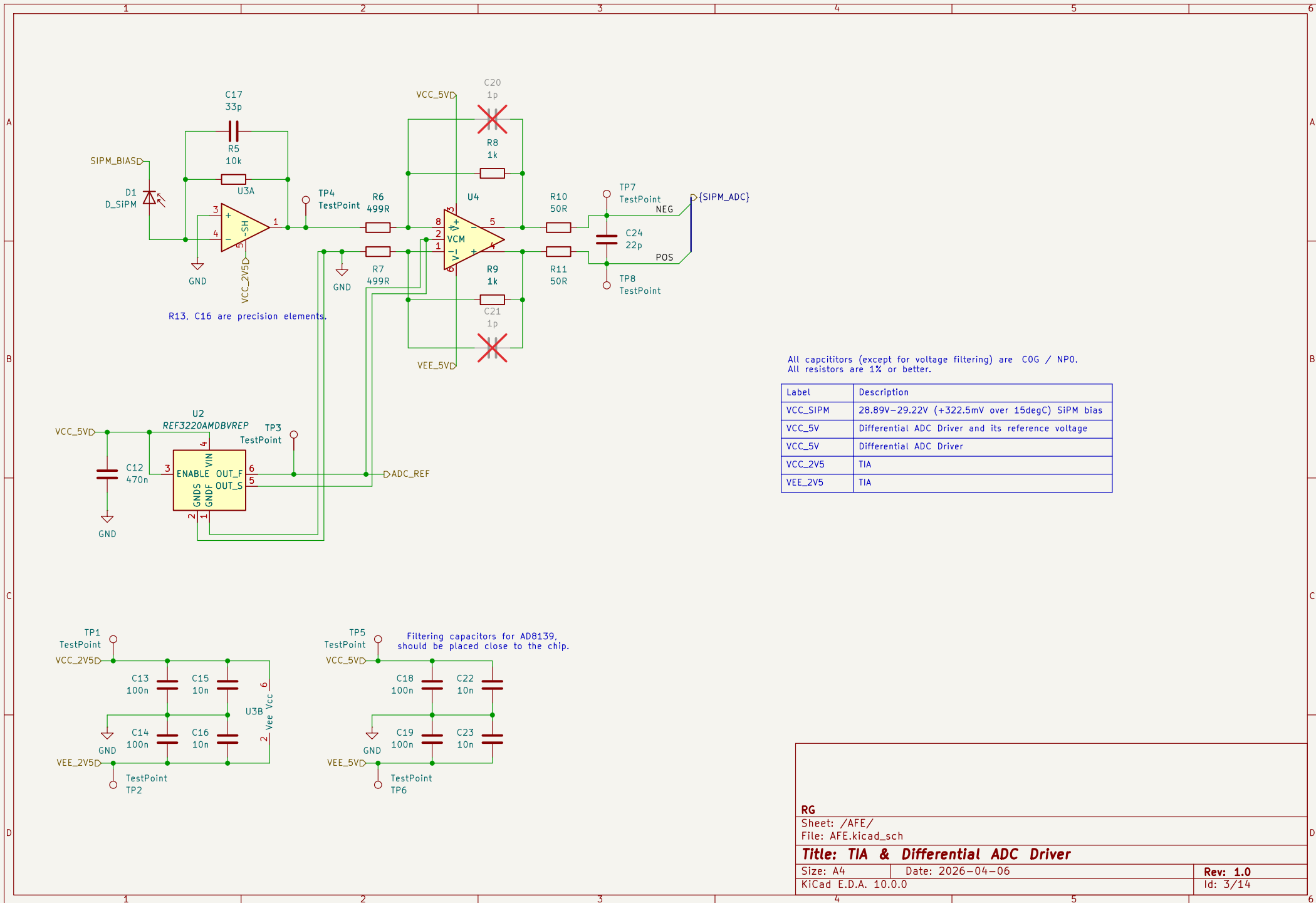
Size: A4

Date: 2026-04-06

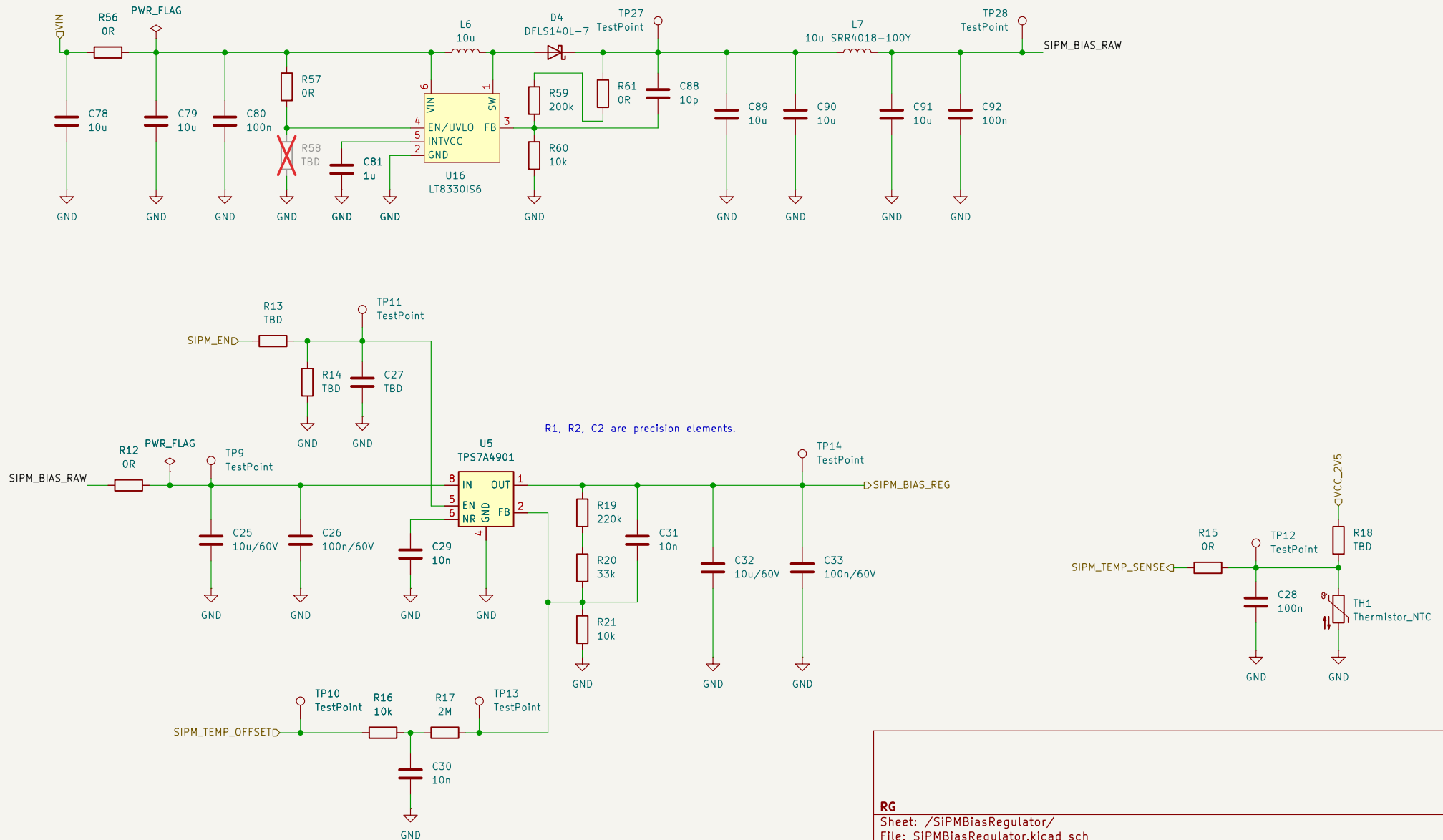
Rev: 1.0

KiCad E.D.A. 10.0.0

Id: 2/14



Schottky should be 40V or more wit low capacitance.
Multiple capacitors in LC filter due to
capacity of capacitor is decreasing at high voltage.



RG

Sheet: /SiPMBiasRegulator/
File: SiPMBiasRegulator.kicad_sch

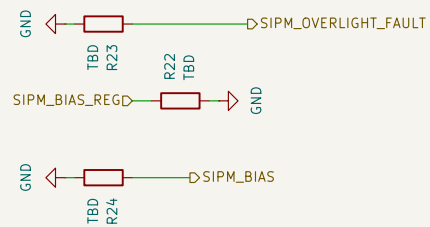
Title: High-Voltage SiPM Bias Generator & Ultra-Low Noise LDO

Size: A4 Date: 2026-04-06

KiCad E.D.A. 10.0.0

Rev: 1.0

Id: 4/14



TODO: this sheet is empty but silence warnings hile working on other sheets

RG

Sheet: /SiPMOverlightProtection/
File: SiPMOverlightProtection.kicad_sch

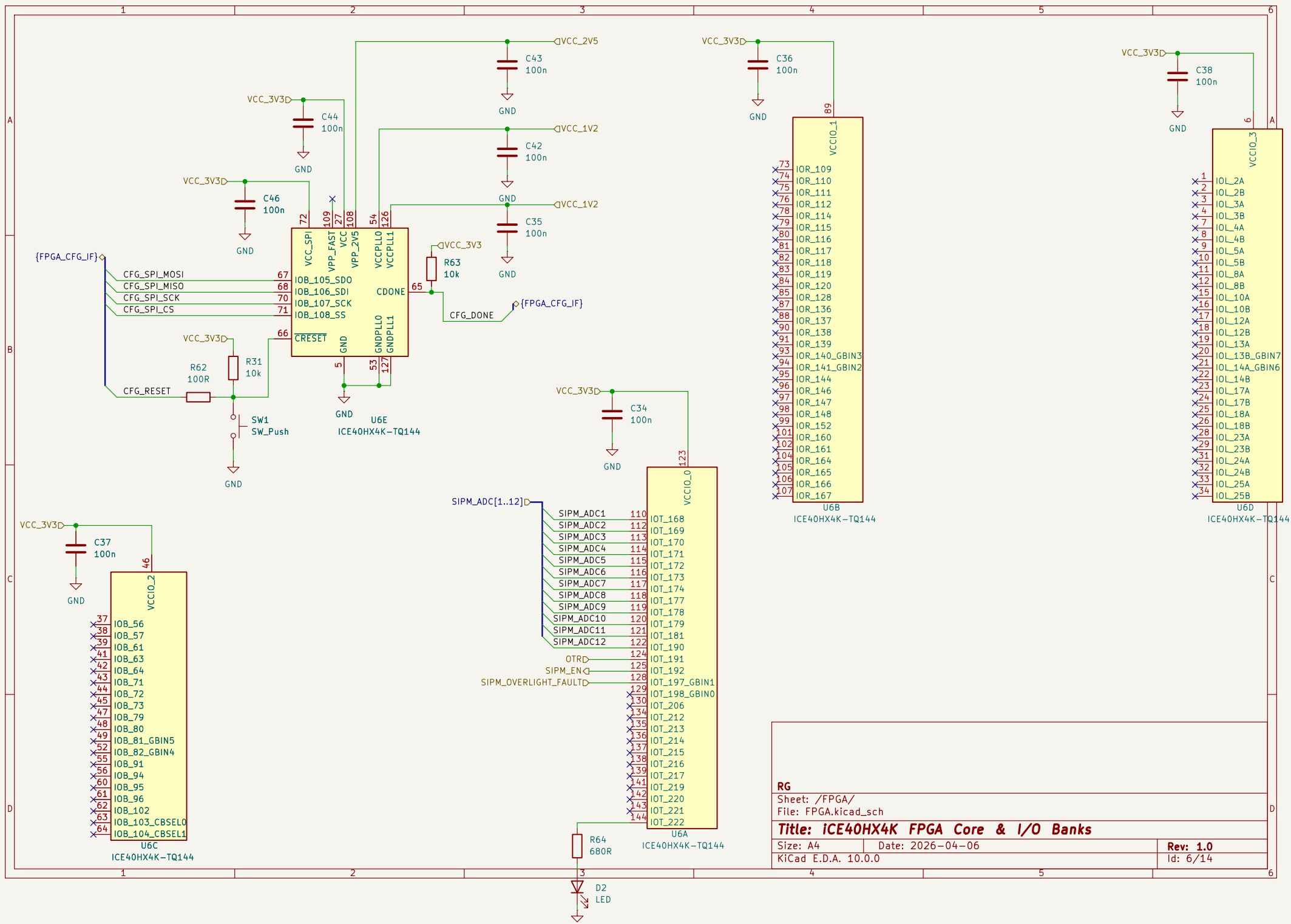
Title: SiPM Over-Illumination Protection

Size: A4	Date: 2026-04-06
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Rev: 1.0

Size: A1	B
KiCad E.D.A. 10.0.0	

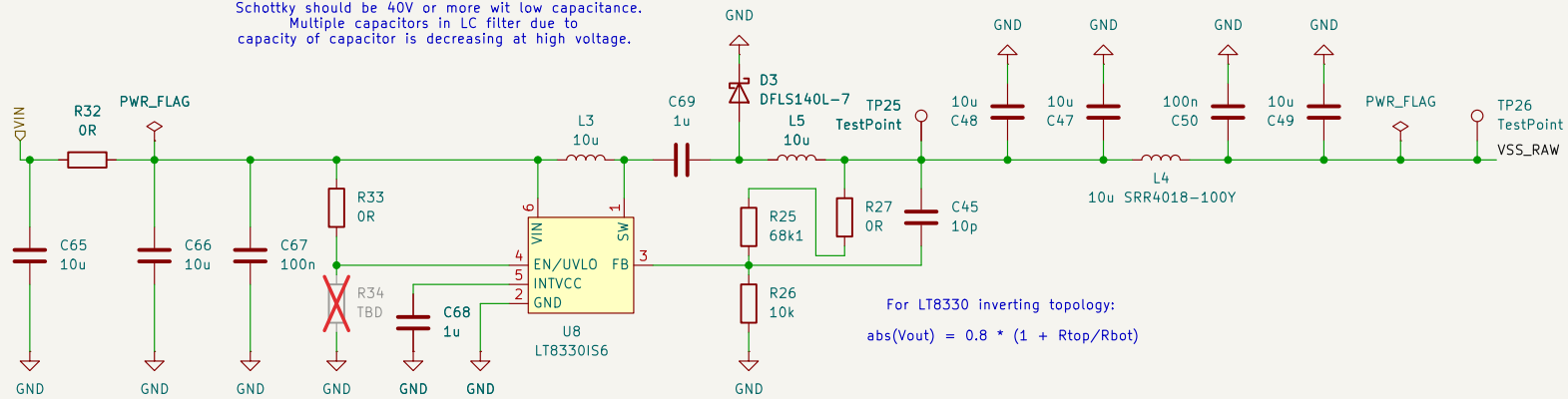
Id: 5/14



1	2	3	4	5	6
A					A
B					B
C					C
D					D
1	2	3	4	5	6

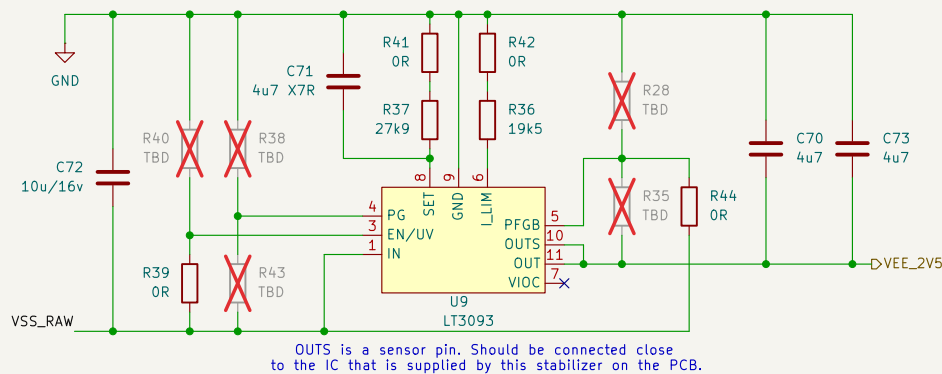
RG		
Sheet: /UserInterface/		
File: UserInterface.kicad_sch		
Title: User Interface		
Size: A4	Date: 2026-04-06	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 7/14

Schottky should be 40V or more wit low capacitance.
Multiple capacitors in LC filter due to
capacity of capacitor is decreasing at high voltage.



For LT8330 inverting topology:
 $abs(V_{out}) = 0.8 * (1 + R_{top}/R_{bot})$

Label	Direction	Description
VSS_RAW	Output	Raw negative voltage (-6.25V)
GND	Bidirectional	Ground



OUTS is a sensor pin. Should be connected close
to the IC that is supplied by this stabilizer on the PCB.

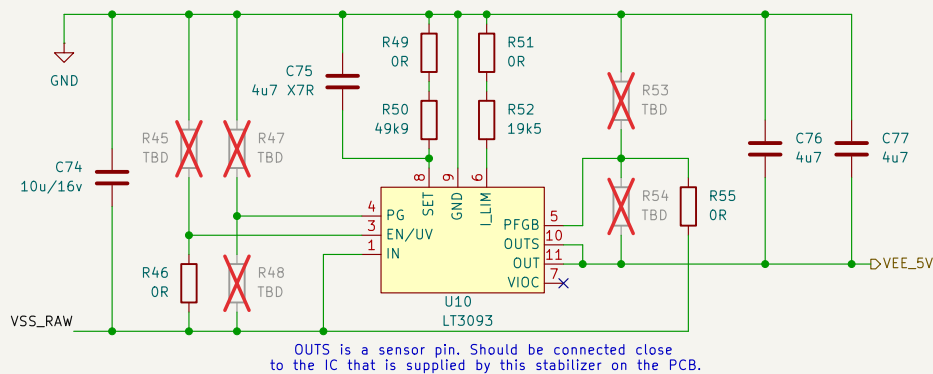
LT3093 calculation:

$$R_{set} = abs(V_{out}) / 100\mu A$$

$$I_{limit} = 1.95k\Omega * A / R_{ilim}$$

for 70 mA (LTC6268): $R_{ilim} = 1.95k / 0.07 = 27.9k$
for 40 mA (AD8139): $R_{ilim} = 1.95k / 0.04 = 48.75k$

PG not used -> leave PG floating
PGFB not used -> tie PGFB to IN
EN/UV tied to IN for always-on operation



OUTS is a sensor pin. Should be connected close
to the IC that is supplied by this stabilizer on the PCB.

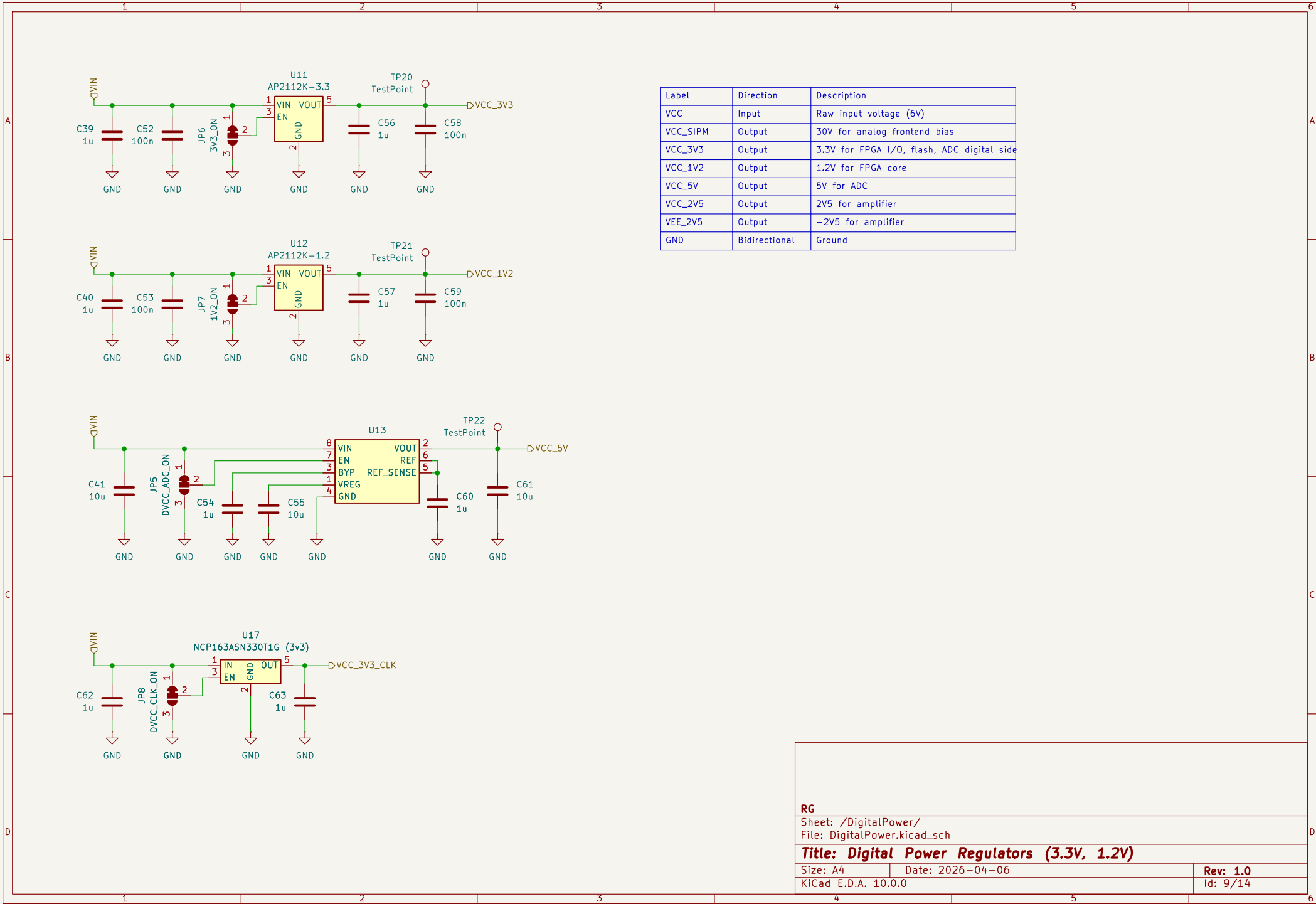
RG

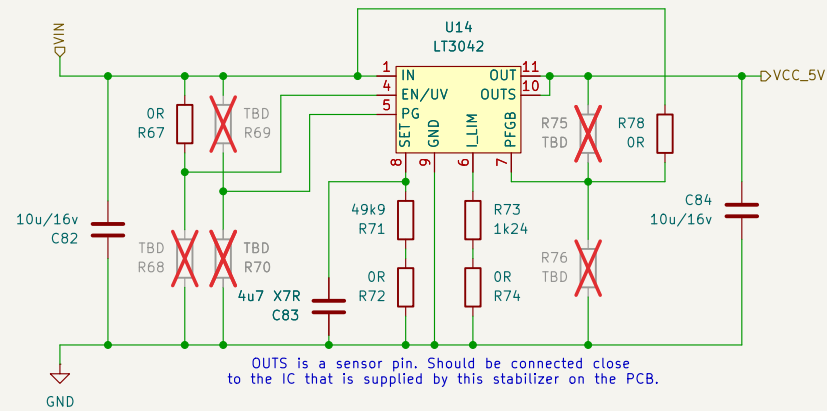
Sheet: /AnalogPowerNegative/
File: AnalogPowerNegative.kicad_sch

Title: Negative Analog Power Supplies (-2.5V, -5V)

Size: A4 Date: 2026-04-06
KiCad E.D.A. 10.0.0

Rev: 1.0
Id: 8/14





LT3042 calculation:

$$R_{set} = \frac{\text{abs}(V_{out})}{100\mu A}$$

$$I_{limit} = \frac{125\text{mA} \cdot k\Omega}{R_{lim}}$$

For:

$$\text{for } 60 \text{ mA: } R_{lim} = \frac{125}{60} = 2.08k$$

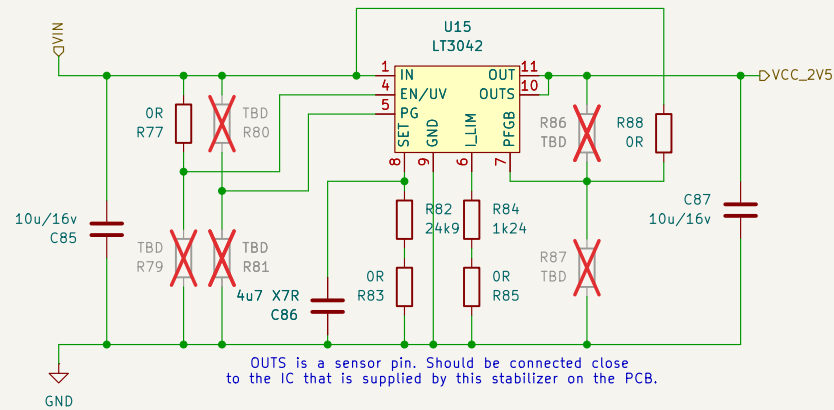
$$\text{for } 70 \text{ mA: } R_{lim} = \frac{125}{70} = 1.79k$$

$$\text{for } 100 \text{ mA: } R_{lim} = \frac{125}{100} = 1.24k$$

PG not used -> leave PG floating

PGFB not used -> tie PGFB to IN

EN/UV tied to IN for always-on operation



RG

Sheet: /AnalogPowerPositive/

File: AnalogPowerPositive.kicad_sch

Title: Positive Analog Power Supplies (2.5V, 5V)

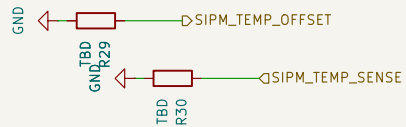
Size: A4

Date: 2026-04-06

Rev: 1.0

KiCad E.D.A. 10.0.0

Id: 10/14



TODO: this sheet is empty but silence warnings hile working on other sheets

RG

Sheet: /SiPMBiasControl/
File: SiPMBiasControl.kicad_sch

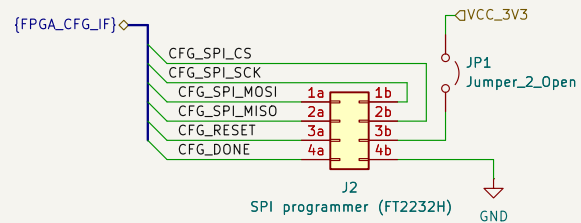
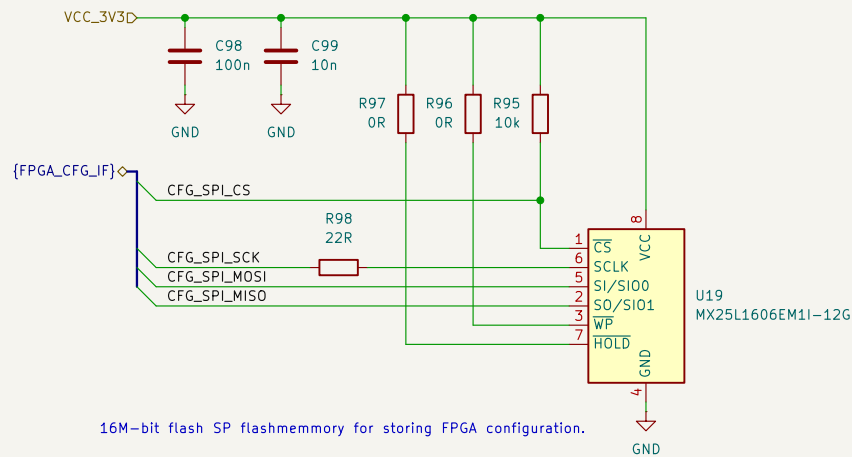
Title: SiPM Bias Control and Temperature Compensation

Size: A4 Date: 2026-04-06

Rev: 1.0

KiCad E.D.A. 10.0.0

Id: 11/14

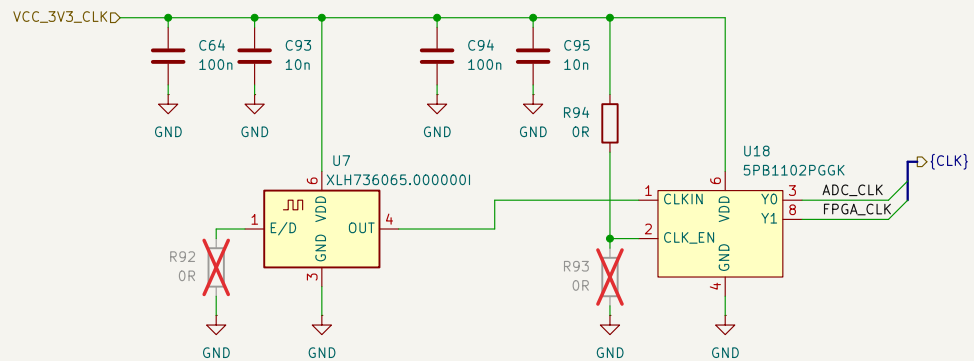


RG

Sheet: /FPGAConfiguration/
File: FPGAConfiguration.kicad_sch

Title: FPGA Configuration Memory – External 16Mb SPI Boot Flash

Size: A4	Date: 2026-04-06	Rev: 1.0
KiCad E.D.A. 10.0.0		Id: 12/14



CLK is 65MHz 25ppm
5PB1102PGGK is 1:2 CLK splitter

RG

Sheet: /ClockGeneration/
File: ClockGeneration.kicad_sch

Title: System Clock – 65 MHz Oscillator and Dual Output Buffer

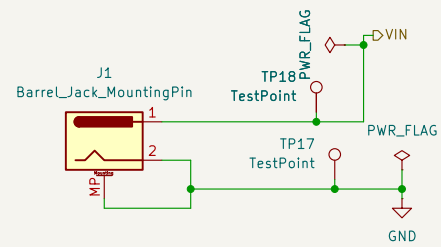
Size: A4 Date: 2026-04-06

Rev: 1.0

KiCad E.D.A. 10.0.0

Id: 13/14

TODO add fuse protection.
THIS IS JUST A DRAFT!!!!



RG

Sheet: /PowerInput/
File: PowerInput.kicad_sch

Title: Barrel Jack Input and Preliminary Protection Interface

Size: A4 Date: 2026-04-06

Rev: 1.0

KiCad E.D.A. 10.0.0

Id: 14/14